

### Problem - 2

1) Given

$$|\psi\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle$$

Required

$$E_F(|\psi\rangle) = ?$$

Solution

For pure state  $|\psi\rangle$ ,  $|\psi\rangle = \sum_i \sqrt{\lambda_i} |\psi_i\rangle |\psi_i\rangle$

Therefore

$$\lambda_1 = \cos^2\theta, \quad \lambda_2 = \sin^2\theta$$

$$\begin{aligned} E_F(|\psi\rangle) &= -\sum_i \lambda_i \log \lambda_i = -\sum_i \lambda_i \log \lambda_i \\ &= -[\cos^2\theta \log_2 \cos^2\theta + \sin^2\theta \log_2 \sin^2\theta] \end{aligned}$$

- Take the angle " $\theta$ " which give smallest and largest value  
At  $E_F(|\psi\rangle) = 0$  and  $E_F(|\psi\rangle) = 1$  for that the  
Entanglement Entropy Vs  $\theta$  graph.

Three are enough for plot

$$\theta = 0, \frac{\pi}{4}, \frac{\pi}{2}$$

At  $\theta = 0$

$$\begin{aligned} \lambda_1 &= (\cos(0))^2 \\ &= (1)^2 \\ &= 1 // \end{aligned}$$

$$\begin{aligned} \lambda_2 &= (\sin(0))^2 \\ &= (0)^2 \\ &= 0 // \end{aligned}$$

At  $\theta = \frac{\pi}{4}$

$$\begin{aligned} \lambda_1 &= (\cos(\frac{\pi}{4}))^2 \\ &= (\frac{1}{\sqrt{2}})^2 \\ &= \frac{1}{2} // \end{aligned}$$

$$\begin{aligned} \lambda_2 &= (\sin(\frac{\pi}{4}))^2 \\ &= (\frac{1}{\sqrt{2}})^2 = \frac{1}{2} \end{aligned}$$

At  $\theta = \frac{\pi}{2}$

$$\begin{aligned} \lambda_1 &= (\cos(\frac{\pi}{2}))^2 \\ &= (0)^2 \\ &= 0 // \end{aligned}$$

$$\begin{aligned} \lambda_2 &= (\sin(\frac{\pi}{2}))^2 \\ &= 1 // \end{aligned}$$

$E_R(\psi) - \text{for } \theta = 0$

$$E_R(\psi) = -[1 \log_2 1 + 0 \log_2 0] = 0 + 0 = 0$$

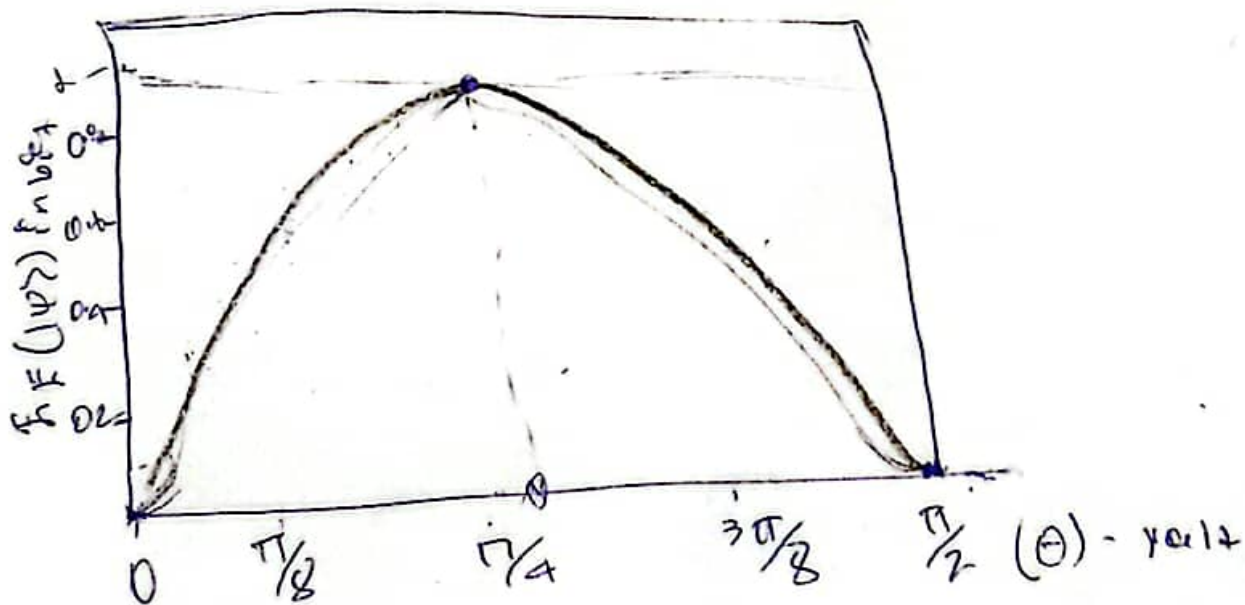
$$\Rightarrow 0 \text{ bit} - \text{minimum value}$$

$E_R(\psi) \text{ for } \theta = \pi/4$

$$E_R(\psi) = -[1/2 \log_2 1/2 + 1/2 \log_2 1/2] = -1/2 \log_2 1/2 = -\log_2 1/2$$

$$= -[\log_2 1 - \log_2 2] = -[0 - 1] = -[-1] = 1$$

$$\Rightarrow 1 \text{ bit} - \text{maximum value}$$



for  $E_R(\psi)$  for  $\pi/2$

$$E_R(\psi) = -[0 \log_2 0 + 1 \log_2 1] = 0 + 0 = 0$$

$$\Rightarrow 0 \text{ bit} - \text{minimum value}$$

2) Given

$$|\psi_1\rangle = \sum_i \sqrt{\lambda_i} |i_{A1}\rangle |i_{B1}\rangle, \quad |\psi_2\rangle = \sum_j \sqrt{\gamma_j} |j_{A2}\rangle |j_{B2}\rangle$$

Required

Show that

$$F_F(|\psi_1\rangle \otimes |\psi_2\rangle) = F_F(|\psi_1\rangle) + F_F(|\psi_2\rangle)$$

Solution

$$\begin{aligned} |\psi_1\rangle \otimes |\psi_2\rangle &= \left( \sum_i \sqrt{\lambda_i} |i_{A1}\rangle |i_{B1}\rangle \right) \otimes \left( \sum_j \sqrt{\gamma_j} |j_{A2}\rangle |j_{B2}\rangle \right) \\ &= \sum_{i,j} \sqrt{\lambda_i \gamma_j} |i_{A1} j_{A2}\rangle |i_{B1} j_{B2}\rangle \end{aligned}$$

$$F_F(|\psi_1\rangle \otimes |\psi_2\rangle) = - \sum_{i,j} \lambda_i \gamma_j \log_2 \lambda_i \gamma_j$$

$$\text{[where, } \log_2 \lambda_i \gamma_j = \log_2 \lambda_i + \log_2 \gamma_j]$$

$$F_F(|\psi_1\rangle \otimes |\psi_2\rangle) = - \sum_{i,j} \lambda_i \gamma_j [\log_2 \lambda_i + \log_2 \gamma_j]$$

$$= - \sum_{i,j} \lambda_i \gamma_j \log_2 \lambda_i - \sum_{i,j} \lambda_i \gamma_j \log_2 \gamma_j$$

$$= - \sum_i \lambda_i \log_2 \lambda_i \underbrace{\sum_j \gamma_j}_{=1} - \sum_j \gamma_j \log_2 \gamma_j \underbrace{\sum_i \lambda_i}_{=1}$$

$$= \underbrace{- \sum_i \lambda_i \log_2 \lambda_i}_{= F_F(|\psi_1\rangle)} - \underbrace{\sum_j \gamma_j \log_2 \gamma_j}_{= F_F(|\psi_2\rangle)}$$



$$\underline{\underline{E_F(1427 \oplus 1427) = E_F(1427) + E_F(1427)}}$$

3) Question

$$|4^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$$

$$S_{AB} = \frac{1}{2}(|00\rangle\langle 00|_{AB} + |11\rangle\langle 11|_{AB})$$

Solution

$$\text{For, } |4^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$$

$$\left. \begin{aligned} p_1 &= \frac{1}{2} \\ p_2 &= \frac{1}{2} \end{aligned} \right\}$$

$$E_F(14^+) = - \left[ \frac{1}{2} \log_2\left(\frac{1}{2}\right) + \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right]$$

$$= -2 \left[ \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right] = -\frac{2}{2} \log_2\left(\frac{1}{2}\right) = -\log_2\left(\frac{1}{2}\right)$$

$$= -[\log_2 1 - \log_2 2] = -[0 - 1] = 1$$

$$= \underline{\underline{1}}$$

where, from,  $S_{AB} = \frac{1}{2}(|00\rangle\langle 00|_{AB} + |11\rangle\langle 11|_{AB})$

$$\left. \begin{aligned} S_A &= \text{Tr}_B \left[ \frac{1}{2}(|00\rangle\langle 00|_{AB} + |11\rangle\langle 11|_{AB}) \right] \\ S_A &= S_B \end{aligned} \right\} = \frac{1}{2} \left[ \text{Tr}_B[|00\rangle\langle 00|_{AB}] + \text{Tr}_B[|11\rangle\langle 11|_{AB}] \right]$$

$$= \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1| = \frac{1}{2} [|0\rangle\langle 0| + |1\rangle\langle 1|]$$

$$= \underline{\underline{\frac{I}{2}}}$$

1)

These calculations-

$$S(\rho_A) = -\text{Tr}[\rho_A \log \rho_A] = -\sum_i p_i \log p_i$$

$$= -\left[\frac{1}{2} \log_2\left(\frac{1}{2}\right) + \frac{1}{2} \log_2\left(\frac{1}{2}\right)\right] = -\frac{2}{2} \log_2\left(\frac{1}{2}\right) = -\log_2\left(\frac{1}{2}\right)$$

$$= -[\log_2(1) - \log_2(2)] = -[-2] = 2$$

$$= 2$$

Similarly,  $S(\rho_B)$

$$S(\rho_B) = -\left[\frac{1}{2} \log_2\left(\frac{1}{2}\right) + \frac{1}{2} \log_2\left(\frac{1}{2}\right)\right] = 2$$

$$= 2$$

So the Entanglement entropy (for pure state) is

$$\boxed{EE(|\phi\rangle) = S(\rho_A) = 2}$$

For the "mixed state" Entropy of  $\rho_A$  and  $\rho_B$

$$\rho_{AB} = \frac{1}{2}(|0\rangle\langle 0|_A \otimes |0\rangle\langle 0|_B) + \frac{1}{2}(|1\rangle\langle 1|_A \otimes |1\rangle\langle 1|_B)$$

So,  $\rho_{AB}$  is - separable mixed state

- No Entanglement

$$EE(\rho_{AB}) = 0$$

- The "hypothetical mixed-state"

Entanglement Entropy

$$EE(\rho_{AB}) \stackrel{?}{=} S(\rho_A)$$

By Corollary,

$$S(SA) = 2$$

$$E_F(SAB) = 0$$

$$E_F(SAB) \stackrel{?}{=} S(SA) = 2$$

### The Argument

⊗ SAB is separable,

$$E_F(SAB) = 0$$

⊗ Therefore entanglement entropy cannot be an entanglement measure for mixed states.

As Given

$$d = 2^m$$

$$|S\rangle = \frac{1}{\sqrt{2}} \sum_i |i\rangle_A |i\rangle_B$$

Solution

$$|i\rangle_A \otimes |j\rangle_B \xrightarrow{\text{LOCC}} |S\rangle$$

from

$$|S\rangle = \frac{1}{\sqrt{2}} \sum_{i=0}^{2^m-1} |i\rangle_A |i\rangle_B$$

$$i \in \{0, \dots, 2^m-1\}$$

$$i = (i_1 i_2 \dots i_m)$$

$$i_k \in \{0, 1\}$$

→  $V_A$  and  $V_B$  maps for LOCC

$$V_A |i\rangle = |i_1\rangle_A \dots |i_m\rangle_A$$

$$V_B |j\rangle = |j_1\rangle_B \dots |j_m\rangle_B$$

Apply on  $|S\rangle$

$$(V_A \otimes V_B) |S\rangle = \frac{1}{\sqrt{2}} \sum_{i=0}^{2^m-1} V_A |i\rangle_A \otimes V_B |i\rangle_B$$

$$(V_A \otimes V_B) |S\rangle = \frac{1}{\sqrt{2}} \sum_{i=0}^{2^m-1} V_A |i\rangle_A \otimes V_B |i\rangle_B$$



$$\begin{aligned}
 (V_A \otimes V_B) |S\rangle &= \frac{1}{\sqrt{2^m}} \sum_{i_1, \dots, i_m} (|i_1 \dots i_m\rangle_A) (|i_1 \dots i_m\rangle_B) \\
 &= \left( \frac{1}{\sqrt{2}} \sum_{i_1=0}^1 |i_1\rangle_A |i_1\rangle_B \right) \otimes \dots \otimes \left( \frac{1}{\sqrt{2}} \sum_{i_m=0}^1 |i_m\rangle_A |i_m\rangle_B \right) \\
 &\quad \quad \quad = |\phi\rangle_2 \quad \quad \quad = |\phi\rangle_m \\
 &= \underbrace{|\phi\rangle_2 \otimes |\phi\rangle_2 \otimes \dots \otimes |\phi\rangle_m}_{= |\phi\rangle_m^{\otimes m}} \\
 &= \underline{\underline{|\phi\rangle_m^{\otimes m}}}
 \end{aligned}$$

Where,

$$\underline{\underline{(V_A^\dagger \otimes V_B^\dagger) |\phi\rangle_m^{\otimes m} = |S\rangle}}$$

- Using only LOCC, Alice and Bob convert shared state into maximally entangled state (Bell pair)
- Convert into each other by LOCC

5) Ques

$$|\Phi_0\rangle = \cos\theta |00\rangle + \sin\theta |11\rangle$$

$$|S\rangle = \frac{1}{\sqrt{3}} (|00\rangle + |11\rangle + |22\rangle)$$

a)  $|\Phi_0\rangle \xrightarrow{\text{by LOCC}} \text{is not Schmidt state}$  7)

Can't increase

Schmidt Rank

If  $0 < \theta < \pi/2$ , then  $\cos \theta \neq 0$ ,  $\sin \theta \neq 0$  so the the  
→ can't decomposition around

$$|\psi\rangle = \cos \theta |00\rangle_{AB} + \sin \theta |11\rangle_{AB}$$

Schmidt Rank = 2

for  $|\psi\rangle$

$$\underline{\underline{SR(|\psi\rangle) = 3}}$$

} has three nonzero Schmidt  
Coefficients  $\frac{1}{\sqrt{3}}$  so, SR=3

⊗ LOCC cannot increase Schmidt rank

Any LOCC protocol is a sequence of local measurements

$$|\psi\rangle \longrightarrow \frac{(A \otimes B)|\psi\rangle}{\| (A \otimes B)|\psi\rangle \|}$$

for some local linear operators  $A, B$

Write  $|\psi\rangle$  in a product state as

$$|\psi\rangle = \sum_{i,j} M_{ij} |i\rangle_A |j\rangle_B$$

$$SR(|\psi\rangle) = \text{rank}(M)$$

After applying  $A \otimes B$ , the new coefficient matrix:

$$M' = A M B^T$$

8)



Matrix Rank over

Q)

$$\text{Rank}(AMB^T) \leq \text{rank}(M)$$

Therefore, On over rank -

$$\text{SR}((A \otimes B) |\psi\rangle) \leq \text{SR}(|\psi\rangle)$$

$|\Phi_0\rangle \xrightarrow{\text{local}} |\Psi_{23}\rangle$  is not possible because  $2 \neq 3$

b) Given

$$\pi_0 = |000\rangle \langle 000|$$

$$\pi_1 = |001\rangle \langle 001| + |010\rangle \langle 010| + |100\rangle \langle 100|$$

$$\pi_2 = |011\rangle \langle 011| + |101\rangle \langle 101| + |110\rangle \langle 110|$$

$$\pi_3 = |111\rangle \langle 111|$$

Solution

let, represent  $\begin{cases} |\Phi_0\rangle = C \\ |\Psi_0\rangle = S \end{cases} \Rightarrow |\Phi_0\rangle = C|00\rangle + S|11\rangle$

$$|\Phi_0\rangle^{\otimes 3} = |\Phi_0\rangle_{A_1 B_1} |\Phi_0\rangle_{A_2 B_2} |\Phi_0\rangle_{A_3 B_3}$$

$$= (C|00\rangle_{A_1 B_1} + S|11\rangle_{A_1 B_1}) (C|00\rangle_{A_2 B_2} + S|11\rangle_{A_2 B_2}) |\Phi_0\rangle_{A_3 B_3}$$

$$= [C^2 |00\rangle_{A_1 A_2} |00\rangle_{B_1 B_2} + CS |01\rangle_{A_1 A_2} |01\rangle_{B_1 B_2} +$$

$$SC |10\rangle_{A_1 A_2} |10\rangle_{B_1 B_2} + S^2 |11\rangle_{A_1 A_2} |11\rangle_{B_1 B_2}] |\Phi_0\rangle_{A_3 B_3}$$

$$= C^3 |000\rangle_{A_1 A_2 A_3} |000\rangle_{B_1 B_2 B_3} + C^2 S |001\rangle_{A_1 A_2 A_3} |001\rangle_{B_1 B_2 B_3} + C^2 S |010\rangle_{A_1 A_2 A_3} |010\rangle_{B_1 B_2 B_3} + C^2 S |011\rangle_{A_1 A_2 A_3} |011\rangle_{B_1 B_2 B_3} +$$

$$+ c^2 s |100\rangle_{A_1 A_2 A_3} |100\rangle_{B_1 B_2 B_3} + c s^2 |101\rangle_{A_1 A_2 A_3} |101\rangle_{B_1 B_2 B_3}$$

$$+ s^2 c |110\rangle_{A_1 A_2 A_3} |110\rangle_{B_1 B_2 B_3} + s^3 |111\rangle_{A_1 A_2 A_3} |111\rangle_{B_1 B_2 B_3}]$$

where,  $A_1 A_2 A_3 = A$  and  $B_1 B_2 B_3 = B$

$$|\psi\rangle = c^3 |100\rangle_A |100\rangle_B$$

$$+ c^2 s [|101\rangle_A |101\rangle_B + |110\rangle_A |110\rangle_B + |100\rangle_A |110\rangle_B]$$

$$+ c s^2 [|101\rangle_A |101\rangle_B + |110\rangle_A |110\rangle_B + |110\rangle_A |110\rangle_B]$$

$$+ s^3 |111\rangle_A |111\rangle_B$$

Apply measurement only for Alice qubit on all four projectors and we will see all the results

$$\textcircled{x} |\psi_0\rangle = (\pi_0 \otimes I) |\psi\rangle$$

$$= (|100\rangle_A \langle 100|_A \otimes I_B) |\psi\rangle$$

$$= c^3 |100\rangle_A |100\rangle_B \quad \text{or } |\psi_0\rangle = |100\rangle_A |100\rangle_B$$

Same for  $\pi_1$ , which is same as above state

$$\textcircled{v} |\psi_1\rangle = (\pi_1 \otimes I_B) |\psi\rangle$$

$$= (|101\rangle_A \langle 101|_A + |110\rangle_A \langle 110|_A) |\psi\rangle$$

$$= c^2 s (|101\rangle_A |101\rangle_B + |110\rangle_A |110\rangle_B + |100\rangle_A |110\rangle_B)$$

$$|\psi_1\rangle = \frac{1}{\sqrt{3}} (|101\rangle + |110\rangle + |100\rangle) \quad |d\rangle$$



$$\overline{|\psi\rangle} = \frac{|\psi\rangle}{\sqrt{P_1}}$$

$$= \frac{c^2 s (|00\rangle|00\rangle + |01\rangle|01\rangle + |10\rangle|10\rangle + |11\rangle|11\rangle)}{\sqrt{1114211}}$$

$$= \frac{c^2 s (|00\rangle|00\rangle + |01\rangle|01\rangle + |10\rangle|10\rangle + |11\rangle|11\rangle)}{\sqrt{3c^4 s^2}}$$

$$= \frac{\cancel{c^2} s}{\cancel{c^2} s \sqrt{3}} (|00\rangle|00\rangle + |01\rangle|01\rangle + |10\rangle|10\rangle + |11\rangle|11\rangle)$$

$$= \frac{(|00\rangle|00\rangle + |01\rangle|01\rangle + |10\rangle|10\rangle + |11\rangle|11\rangle)}{\sqrt{3}}$$

For,  $|\psi_2\rangle = (\pi_2 \otimes I_B) |\psi\rangle$

$$= (|0\rangle|0\rangle|0\rangle + |1\rangle|0\rangle|1\rangle + |0\rangle|1\rangle|0\rangle + |1\rangle|1\rangle|1\rangle) |\psi\rangle$$

$$= c s^2 (|00\rangle|00\rangle + |01\rangle|01\rangle + |10\rangle|10\rangle + |11\rangle|11\rangle)$$

$$P_2 = \langle \psi_2 | \psi_2 \rangle = 3 c^2 s^4$$

$$\overline{|\psi_2\rangle} = \frac{|\psi_2\rangle}{\sqrt{P_2}} = \frac{1}{\sqrt{3}} (|00\rangle|00\rangle + |01\rangle|01\rangle + |10\rangle|10\rangle + |11\rangle|11\rangle)$$

For both  $\pi_1$  and  $\pi_2$  - projector I have  $S_{\pi_1} = 3$  and



For, 3

$$|\psi\rangle = (\pi_3 \otimes I) |\psi\rangle$$
$$= 5^3 |xxx\rangle_A |xxx\rangle_B$$

$$P_3 = ||\psi_3\rangle|^2 = 6^6$$

$$|\psi_3\rangle = \frac{|\psi\rangle}{\sqrt{P_3}} = \frac{|xxx\rangle_A |xxx\rangle_B}{\sqrt{6^6}}$$

$$S_R(|\psi_3\rangle) = 2$$

$$S_R(|\psi_2\rangle) = S_R(|\psi_2\rangle) = 3$$

$$S_R(|\psi_0\rangle) = 2$$

There for

⊗ from the above all calculation on Alice measurement fails at

At, 0 and 3

$$- S_R(|\psi_0\rangle) = S_R(|\psi_3\rangle) = 4$$

- the state become separable

⊗ from the measurement is Alice measurement I have give a result for projector error

$\pi_1$  and  $\pi_2$

$$- S_R(|\psi_1\rangle) = 3$$

$$- S_R(|\psi_2\rangle) = 3$$

the state  
are entangled

## Succeds

- for norm 2 and 2
- Schmidt rank = 3
- Entangled state

## Pauls

- for norm 0 and 3
- Schmidt rank = 1
- separable state

## 6) Solution

ⓐ Let  $|\psi\rangle$  be a bipartite pure state with Schmidt decomposition

$$|\psi\rangle = \sum_i \sqrt{\lambda_i} |i\rangle_A |i\rangle_B$$

ⓑ Entanglement Entropy

$$E_E(|\psi\rangle) = S(\rho_A) = S(\rho_B) = -\sum_i \lambda_i \log \lambda_i$$

h) Entanglement Concentration  $|\psi\rangle^{\otimes n} \rightarrow \text{Bell pairs}$

Consider  $n$ -copies,

$$|\psi\rangle^{\otimes n} = \sum_{i^n} \sqrt{\lambda_{i^n}} |i^n\rangle_A |i^n\rangle_B$$

Alice and Bob can classical communication

$$K \approx n E_E(|\psi\rangle)$$

$$K \approx n E_E(|\psi\rangle)$$

Entanglement dilution: Bell pairs  $\rightarrow |\psi\rangle^{\otimes n}$

Given,  $k$  EPR-pairs Alice and Bob can create  $|\Phi\rangle^m$  by LOCC with vanishing error

$$k \geq m E_E(|\Phi\rangle)$$

Combine them to convert

$$|\Psi\rangle^{\otimes n} \rightarrow |\Psi\rangle^{\otimes m}$$

Chain the two steps

$$|\Psi\rangle^{\otimes n} \xrightarrow{\text{concentration}} |\Phi\rangle^{\otimes m} \xrightarrow{\text{dilution}} |\Psi\rangle^{\otimes n}$$

$$m E_E(|\Phi\rangle) \leq n E_E(|\Psi\rangle)$$

Optimal Asymptotic Conversion Rate

$$\frac{m}{n} \leq \frac{E(|\Psi\rangle)}{E(|\Phi\rangle)}, \quad \text{error} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Optimal (converse)

$$\left. \begin{aligned} E(|\Psi\rangle^{\otimes n}) &= n E_E(|\Psi\rangle) \\ E(|\Phi\rangle^{\otimes m}) &= m E_E(|\Phi\rangle) \end{aligned} \right\} : n E_E(|\Psi\rangle) \geq m E_E(|\Phi\rangle)$$

$$\lim_{n \rightarrow \infty} \frac{m}{n} = \frac{E(|\Psi\rangle)}{E(|\Phi\rangle)} = \frac{S(\text{Tr}_B(|\Psi\rangle\langle\Psi|))}{S(\text{Tr}_B(|\Phi\rangle\langle\Phi|))}$$

(4)